

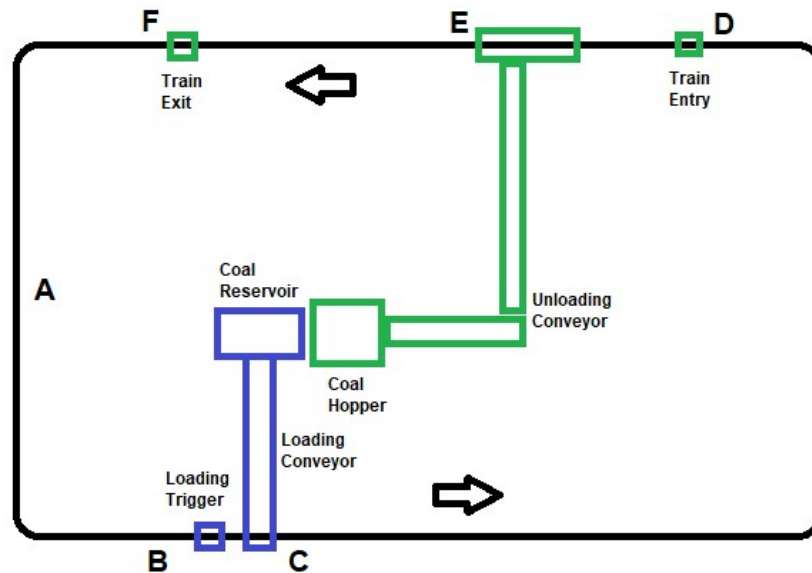
# **Coal Hopper**

## **Design Notes**

**Version 1.0**

## Overview

The Coal Wagon exhibition layout of the Bracknell Railway Society runs a continuous loop of trains, pulling several coal wagons, that are filled and emptied automatically. A simplified layout is shown below.



The train, pulling a number of coal wagons, travels anti-clockwise starting empty at point A. Each wagon has a small magnet fitted and when it passed over the loading trigger reed switch at point B, it causes the train to stop after a short delay, with the wagon under the loading chute C. This also starts the loading conveyor to move coal from the reservoir into a loading bucket. When this fills to a certain amount, the bucket tips and dispenses the coal down the chute into the wagon. A tilt switch on the bucket stops the conveyor and after a short delay, restarts the train. As each wagon passes the loading trigger, the process repeats. When all the wagons are loaded, the train continues around the loop.

When the train reaches train entry point D, the unloading conveyor is started. As the wagons reach point E, the coal is automatically unloaded through the bottom of the wagon. The coal is moved up the unloading conveyors to the coal hopper.

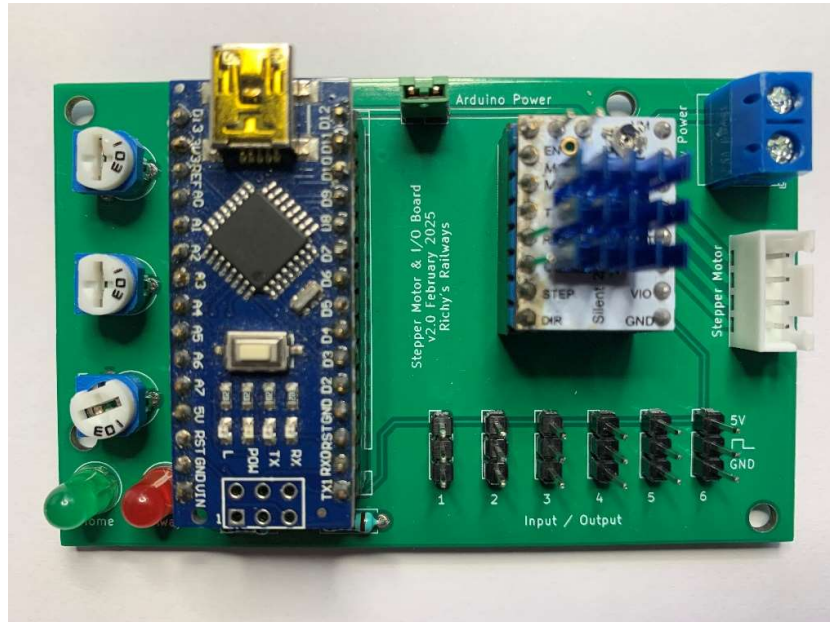
Once the train reaches the train exit point F, and after a short delay, the unloading conveyor is turned off. The coal hopper is raised up the tower and is tipped into the coal reservoir. It then returns to the base.

The train reaches point A again and the cycle repeats itself.

This document covers the green unloading part of the layout and how the reservoir of coal in the loading tower is replenished.

## Hardware

A custom PCB has been designed, based on an Arduino microcontroller that detects the train location and moves a coal hopper lift using a stepper motor. The final version of the board is shown below.



The main part of the board is the Arduino Nano. The USB interface on this is used only for programming and debugging. In normal operation, it takes power from the 12V supply used for the Stepper Motor.

The jumper to the right of the Arduino can be used to disconnect the 12V supply. When the Arduino is powered through its USB connector, the jumper should be removed to avoid power supply conflicts and protect the host computer. In normal operation with no USB connection, the jumper should be fitted.

The stepper motor driver is a Silent\_2209 board. This drives the stepper motor in 1/8 micro steps with very little noise compared to a normal driver board.

The 3 variable resistors or trimmers are used to tune the configuration of the board. These are:

- Delay – the time to wait before moving to the home or away position.
- Speed – How fast the stepper motor turns.
- Length – The number of turns the motor will take to move from home to away.

There are 6 I/O interfaces on the board. These are used as follows:

1. Home switch (input) – This is connected to a 'normally-open' microswitch.
2. Away switch (input) – This is currently not fitted.
3. Conveyor motor control (output) – This drives a relay board.
4. Train entry sensor (input) – This is a small reed switch.
5. Train exit sensor (input) – This is a small reed switch.
6. Busy / Track power (output) – This drives a relay board.

The two LEDs display different patterns depending on what state the board is in.

The Green LED shows the 'Home' state, and the Red LED shows the 'Away' state. These will light and flash in different combinations to show what function the board is performing.

All the above will be covered in detail within the document.

## Software Description

The Arduino code is written in 'C' and can be found in the GitHub repository at <https://github.com/lucari/Coal-Hopper>, along with the PCB design files. The code should be used with the following description to understand how it works.

The software design is based on a finite-state-machine model. The code will be in one state until some trigger causes it to perform an action and move to a new state. The possible states are: Initialise, Home, Move-Away, Away, Move-Home and Error.

### Initialise

Once the Arduino starts up and has performed its local setup functions, it is in the Initialise state (InitialiseHopper function). It flashes the LEDs to show test they are working and reads the trimmer pots to get the tuneable values.

Because the power could have been removed at any time, the location of the hopper is checked. If the home switch is active (hopper is home), then it is moved away for a number of turns and then returned home. If the Home switch is inactive (hopper is away), the hopper is returned home. This checks that the home switch is working correctly. If the state of the home switch does not change after a number of turns, the program enters the **Error** state. If the home switch does change correctly, the program enters the **Home** state.

### Home

This is the state that the program will be in most of the time (WaitForHopperTrigger function). The stepper motor is disabled during this time to save power. While it is idle, it will check the trimmer pots every 15 seconds and update the tuneable values.

The function is waiting for one of two triggers to occur. The 'Train Entry' trigger (trainSensorPin) going low cause the unloading conveyors to start. This is handled by taking the conveyorMotorPin pin low, which is connected to an external relay board. It also starts a 60 second time that is checked within the loop. When this expires, the pin is raised and stopping the conveyors. The state is not changed by this event.

When the 'Train Exit' trigger (bucketFullSensorPin) goes low, the idle loop is exited. The program will wait for the number of seconds defined by the trim pots. This is to ensure that the unloading conveyors are empty and the train has cleared the section. The track power is then turned off by taking the hopperBusyPin pin low. This is connected to an external relay board. It also stops the unloading conveyors if they were running. The program enters the **Move-Away** state.

### **Move-Away**

This is the transient state where the hopper is being moved from the home (lower) to the away (raised) position (MoveHopperToAway function). The stepper motor board is driven by setting the direction pin high and the enable pin low. The step pin is pulsed by setting it high, wait, low, and wait. The wait time is set by the trimmer pots, and the optimum value was found to be 200uS. At the start and end of the run, the delay is doubled to slow it down.

It was found that a steady stream of pulses works best. When the debug messages were included, the stepper produced a knocking sound at each turn. So, these have been commented out.

The stepper motor is moved until the away switch is reached (if fitted) or the number of turns specified by the trim pots is reached. The number of turns required on the current setup is 53. The driver board is set for 1/8 steps so a full turn will require  $8 * 200 = 1600$  pulses.

When the endpoint is reached, the stepper motor is disabled to save power. The program enters the **Away** state.

### **Away**

This is the transient state when the hopper is at the highest point and is emptying the coal into the reservoir (WaitForHopperToEmpty function). It simply waits for the number of seconds defined by the trimmer pots. The value of 10 seconds seems to work best.

Once the timer expires, the program enters the **Move-Home** state.

### **Move-Home**

This is the transient state where the hopper is being moved from the away (raised) to the home (lower) position (MoveHopperToHome function). It operates in the same manner as the Move-Away state, except the direction pin is set low. The Home switch will always be fitted so this is the trigger that will stop the hopper moving. Once reached, the program enters the **Home** state.

If the number of turns is exceeded, the program enters the **Error** state.

### **Error**

When something unexpected happens, the program enters the Error state and halts processing.

## LED Patterns / Debugging

The two LEDs show the current state and functions of the program.

| Home<br>Green   | Away<br>Red      | Meaning                                                     |
|-----------------|------------------|-------------------------------------------------------------|
| On<br>On<br>Off | On<br>Off<br>Off | Initialise / Start up sequence                              |
| On<br>Off       | Off<br>On        | Error state. Alternates every second                        |
| On              | Off              | Home state – waiting for trigger                            |
| On              | Fast             | Home state – delay before moving away (fast flash Away LED) |
| On              | Flash            | Move-Away state – alternate Away LED each turn              |
| Off             | On               | Away state – waiting for timeout                            |
| Flash           | On               | Move-Home – alternate Home LED each turn                    |

The USB port of the Arduino can also be used to view debug messages from the program. The Arduino Power jumper should be removed from the board to avoid conflicts. A USB lead should be connected to a computer and the Arduino IDE started. The Serial Monitor can be found under the Tools menu. The serial port speed should be set to 115200.

```
InitialiseHopper: Start
InitialiseHopper: Hopper away from home position. Moving to
home.
InitialiseHopper: Full turn home complete.
InitialiseHopper: Full turn home complete.
InitialiseHopper: Full turn home complete.
InitialiseHopper: Hopper successfully moved home.
WaitForHopperTrigger: At start, waiting 5 seconds.
WaitForHopperTrigger: Waiting for trigger to move.
WaitForHopperTrigger: Setting defaultDelay (Speed) to 206
WaitForHopperTrigger: Setting turnsToAway (Length) to 53
WaitForHopperTrigger: Setting awayWaitTime (Delay) to 10146
```

## Connectivity

The connections to the board are shown below. It should be noted that the 5V supply is fed from the Arduino on-board regulator and can only provide limited current. It cannot be used to drive relay boards or servos. The signal side (opto isolator) connection is ok to use 5V, but not the relay side (JD-VCC).

These connections are directly connected.

|       |     |       |                                             |
|-------|-----|-------|---------------------------------------------|
| I/O 1 | Gnd | Black | Home switch direct on green and black wire. |
|       | Sig | Green |                                             |
|       | +5V | -     |                                             |
| I/O 2 | Gnd | -     | Not used                                    |
|       | Sig | -     |                                             |
|       | +5V | -     |                                             |

These connections are through the patch block.

|       |     |        |                                                                                                                                                   |
|-------|-----|--------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| I/O 3 | Gnd | -      | Unload conveyor relay board. Just uses +5V and signal. Relay board must use external power for relay side (JD-VCC)                                |
|       | Sig | Brown  |                                                                                                                                                   |
|       | +5V | Red    |                                                                                                                                                   |
| I/O 4 | Gnd | Orange | Reed switch. Just uses Sig and Gnd.                                                                                                               |
|       | Sig | Yellow |                                                                                                                                                   |
|       | +5V | Green  |                                                                                                                                                   |
| I/O 5 | Gnd | Blue   | Reed switch. Just uses Sig and Gnd.                                                                                                               |
|       | Sig | Violet |                                                                                                                                                   |
|       | +5V | Grey   |                                                                                                                                                   |
| I/O 6 | Gnd | -      | Track Power isolator relay board. Just uses +5V and signal. +5V is common with I/O 3. Relay board must use external power for relay side (JD-VCC) |
|       | Sig | White  |                                                                                                                                                   |
|       | +5V | Black  |                                                                                                                                                   |



## Stepper Motor Wiring

It has been found that there are two type of stepper motor interfaces and require different cables. If the wrong cable is used, the stepper motor will judder and not turn.

| 4 pin (driver) | 6 pin type 1 | 6 pin type 2 |
|----------------|--------------|--------------|
|                | A2           | A2           |
| A2             | -            | -            |
| A1             | A1           | B1           |
| B1             | B1           | A1           |
| B2             | -            | -            |
|                | B2           | B2           |

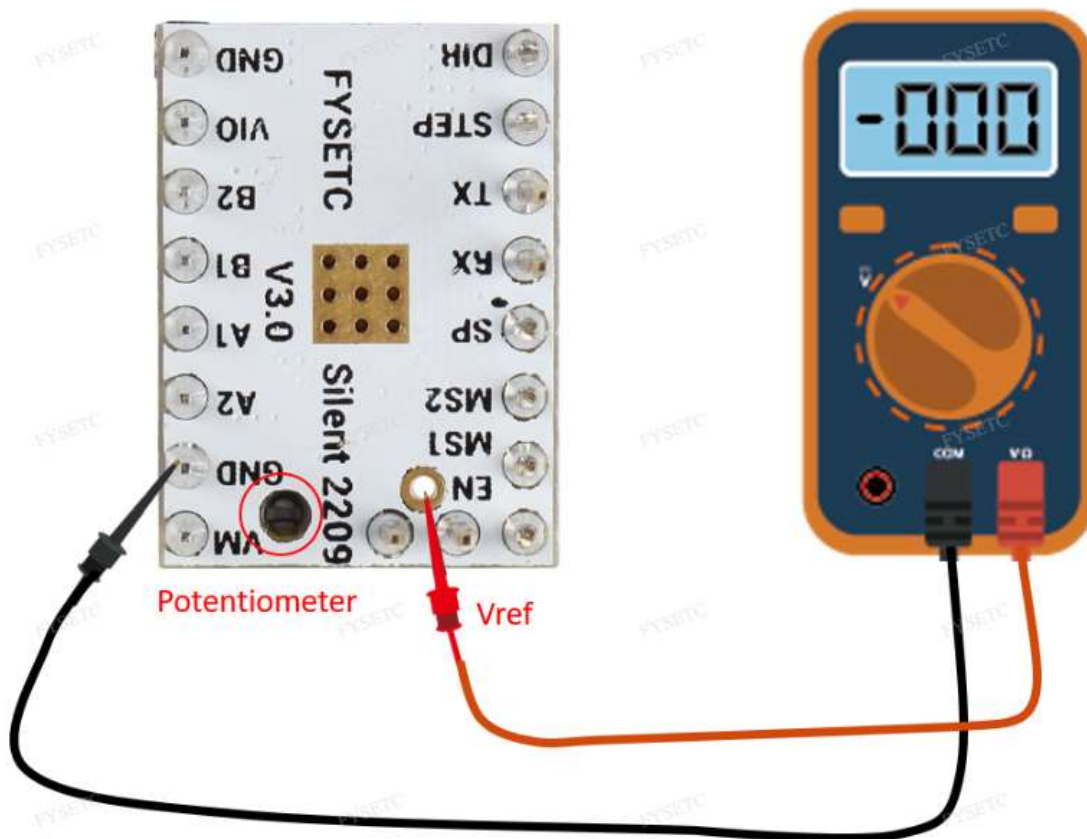
The pins on the stepper motor should be checked for connectivity to determine the correct cable.

\*\*\* The stepper motor type in the current installation is unknown \*\*\*

## Stepper Motor Current Limit

The current limit must be set on the stepper driver board, or it will consume high current and could impact the wiring. The current is set by applying power to the VM and Ground pins and then measuring the voltage on the Vref pin. The current limit can be adjusted using the trim pot. The stepper motor **must not be** connected to the board, or it will cause damage.

The current is defined as  $V_{ref} * 0.71$ . The board is set to 0.6A with a Vref of 0.85V



The weblink <https://wiki.fysetc.com/docs/Silent2209> gives full details of the stepper motor driver board.